

Agenda

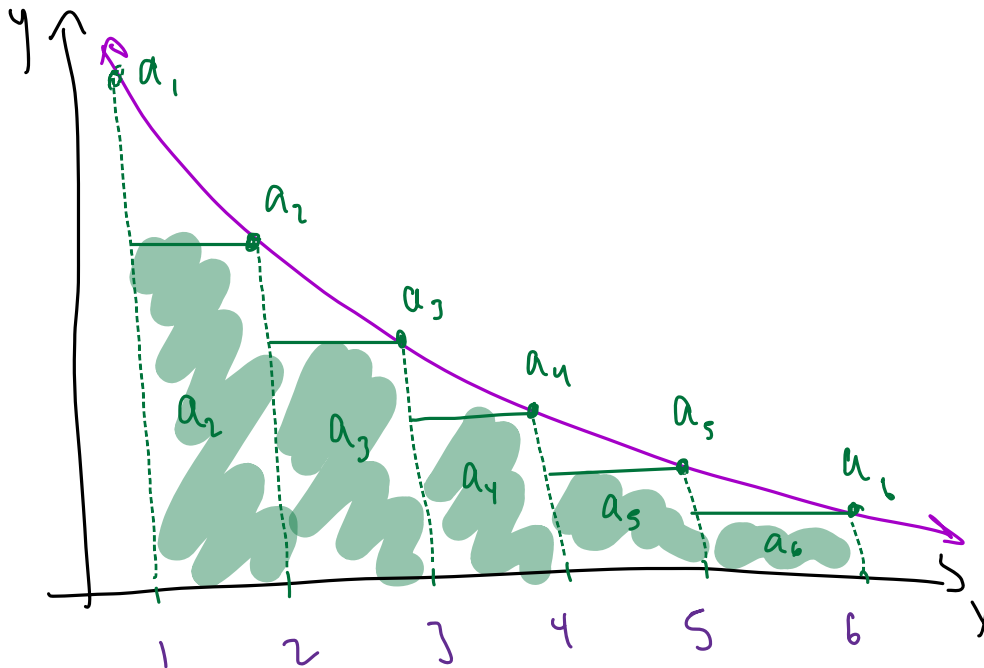
1. Announcements

- Exam 1 regrade requests due Friday 3/24 at 11:59pm
- 9.5 and 11.1 HW due tonight
- Discussion Quiz 6 tomorrow over 
- Mid-semester feedback: go.wisc.edu/udahj3
- I won't have office hours today

2. Section 11.3

The Integral Test *(the last resort)*

Take a function $f(x) > 0$ that is decreasing on $[1, \infty)$
 and $f(n) = a_n$ for all $n \geq 1$



Notice $0 < \text{total area of rectangles} = \sum_{n=2}^{\infty} a_n \leq \int_1^{\infty} f(x) dx$

So we can conclude,

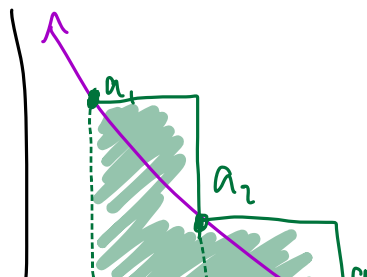
if $\int_1^{\infty} f(x) dx$ converges,

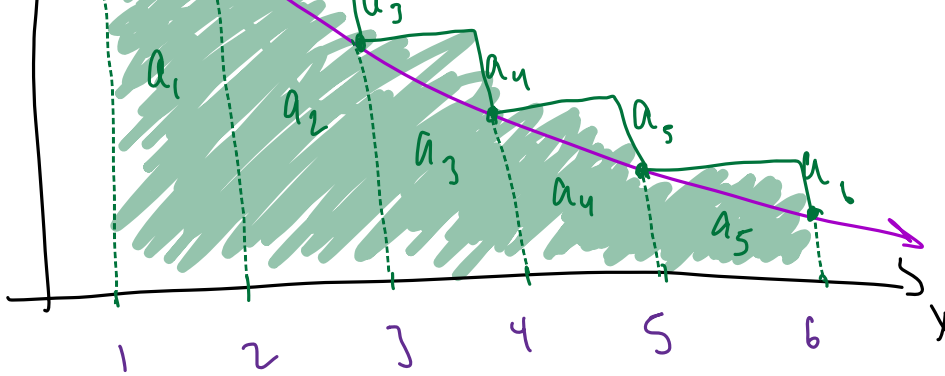
then $\sum_{n=2}^{\infty} a_n$ also converges.

and also $(a_1 + \sum_{n=2}^{\infty} a_n = \sum_{n=1}^{\infty} a_n)$ converges

can add finite things to series
 will not change convergence.

Divergence





Notice total area of rectangles $\sim \sum_{n=1}^{\infty} a_n \geq \int_1^{\infty} f(x) dx \geq 0$

If $\int_1^{\infty} f(x) dx$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges

(the last resort test)

The Integral Test: Suppose f is a positive, decreasing ^{$\frac{d}{dx}[f(x)] < 0$} function on $[1, \infty)$, where $f(n) = a_n$ for all n .

- 1.) If $\int_1^{\infty} f(x) dx$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.
- 2.) If $\int_1^{\infty} f(x) dx$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.

Example: Does $\sum_{n=3}^{\infty} \frac{1}{n^2}$ converge or diverge? *Can start at any index*

Define $f(x) = \frac{\ln x}{x}$ on $[3, \infty)$

is $f(x) > 0$ on $[3, \infty)$? yes. $\ln x > 0$, $x > 0$ *must justify (simply)*

is $f(x)$ decreasing? $\frac{1 - \ln x}{x^2}$ yes.

enough to show $1 - \ln x < 0$ on $[3, \infty)$

$$\begin{array}{l} 1 < \ln x \\ 1 < x \\ e < 3 \end{array} \checkmark$$

$$\begin{aligned} \int_3^{\infty} \frac{\ln x}{x} dx &= \lim_{t \rightarrow \infty} \int_3^t \frac{\ln x}{x} dx = \lim_{t \rightarrow \infty} \left[\frac{\ln x^2}{2} \right]_3^t \\ &= \lim_{t \rightarrow \infty} \left[\frac{1}{2} (\ln t)^2 - \frac{1}{2} (\ln 3)^2 \right] = \infty, \\ &\text{diverges} \end{aligned}$$

Therefore $\sum_{n=3}^{\infty} \frac{\ln n}{n}$ diverges by the integral test.

Always end q's with a complete sentence.

Activity: Does $\sum_{n=1}^{\infty} e^{-n}$ converge or diverge?

$f(x) = e^{-x}$ where $e^{-x} > 0$

$\frac{d}{dx} [e^{-x}] = -e^{-x}$ which is decreasing everywhere

$$\int_1^{\infty} e^{-x} dx = \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx = \lim_{t \rightarrow \infty} \left[e^{-x} \right]_1^t$$

$$= -e^{-t} + e^{-1}$$

$$= -\frac{1}{e^t} + \frac{1}{e}$$

$$= \left(\frac{1}{e} \right) \rightarrow \text{converges}$$

Therefore, $\sum_{n=1}^{\infty} e^{-n}$ converges via the Integral Test.

p-Series

Recall: $\int_1^{\infty} \frac{1}{x^p} dx$ converges $p > 1$
diverges $p \leq 1$

Theorem: Let p be a constant.

1.) If $p > 1$, then $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges

2.) If $p \leq 1$, then $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges
/

Activity: Determine if the series converges or diverges.

1.) $\sum_{n=1}^{\infty} \frac{1}{n^2}$

converges
via p-series
with $p > 1$

2.) $\sum_{n=1}^{\infty} \frac{1}{n^{1.01}}$

converges
via p-series
with $p > 1$

3.) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

diverges
via p-series
with $p \leq 1$

4.) $\sum_{n=15}^{\infty} \frac{1}{n}$

diverges
via p-series
with $p \leq 1$

Estimating Sums Using Improper Integrals

How many terms do we have to add to be "close enough" to estimate what a series converges to?

given $S = \sum_{n=1}^{\infty} a_n$ and the partial sum $S_n = a_1 + a_2 + \dots + a_n$,

we define the remainder $R_n = a_{n+1} + a_{n+2} + \dots$

then $S = \sum_{k=1}^n a_k + \sum_{k=n+1}^{\infty} a_k = S_n + R_n$, so $S - S_n = R_n$

Remainder Estimation: Suppose $f(k) = a_k$ where f is a positive, decreasing function for all $x \geq n$. If $\sum_{k=1}^{\infty} a_k$ converges, then
(If the integral test applies and it converges.)

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$$

$$\int_{n+1}^{\infty} f(x) dx \leq S - S_n \leq \int_n^{\infty} f(x) dx$$

$$S_n + \int_{n+1}^{\infty} f(x) dx \leq S \leq S_n + \int_n^{\infty} f(x) dx$$

so we can bound S by things we can calculate.

Example: Estimate $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using $S_{10} = \frac{1}{1} + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{100} \approx 1.5498$

Activity: Identify the appropriate tool to determine if the Series converges or diverges (geometric series, telescoping series, divergence test, p-series, integral test).

1.) $\sum_{n=3}^{\infty} \frac{n^2+5}{n^2+2n-3}$

2.) $\sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n^4} \right)$

3.) $\sum_{n=4}^{\infty} \frac{6}{5^n}$

4.) $\sum_{n=0}^{\infty} \frac{1}{3n+5}$

5.) $\sum_{n=2}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right)$

6.) $\sum_{n=0}^{\infty} 2^n$